

CHAN YI HENG

+65 8862 0913 | c@chanyh.com | chanyh.com | cyh-chan | chanyiheng

EDUCATION

National University of Singapore

Master of Science in Statistics

Singapore, SG

Aug 2026 – Expected 2027

Singapore University of Social Sciences

Bachelor of Science (Honours) in Mathematics

Singapore, SG

Jan 2022 – Dec 2025

Singapore Polytechnic

Diploma in Banking & Finance

Singapore, SG

Apr 2016 – Mar 2019

EXPERIENCE

Teaching Assistant

Singapore University of Social Sciences

Jan 2026 – Apr 2026

Singapore, SG

- Led weekly tutorials for approximately 30 undergraduate students in Statistical Analysis, covering probability theory, statistical inference, and linear regression.
- Developed a structured revision framework synthesizing core theoretical principles to reinforce formal statistical reasoning.
- Emphasized rigorous model formulation and assumption diagnostics, cultivating disciplined quantitative problem-solving practices.

Research Assistant

Singapore University of Social Sciences, University Research Office

Oct 2025 – Present

Singapore, SG

- Collaborated on institutional research initiatives, overseeing data preprocessing, validation, and analytical implementation.
- Maintained reproducible data integrity and experimental protocols for large-scale psychological datasets.
- Synthesized statistical findings into formal technical documentation for principal investigators and academic review.

Research Intern

A*STAR, Advanced Remanufacturing and Technology Centre

Nov 2024 – Aug 2025

Singapore, SG

- Designed deep learning time-series models for FMCG demand forecasting, improving inventory allocation under uncertainty.
- Developed an Explainable AI framework leveraging SHAP to quantify feature attribution and validate model consistency.
- Reproduced and extended contemporary architectures to strengthen empirical robustness and comparative benchmarking.

Finance Operations Analyst

Indeed

Sep 2021 – Dec 2021

Singapore, SG

- Executed cross-border financial data migrations with strict compliance to international reporting standards.
- Optimized operational workflows and data validation processes within Salesforce environments.

Finance Specialist

Singapore Armed Forces

Aug 2019 – Dec 2021

Singapore, SG

- Managed financial operations and budget allocations under central government regulatory frameworks.
- Conducted internal audits and risk assessments, reporting governance findings to senior leadership.
- Administered operational budgets during COVID-19 deployments with full compliance and transactional accuracy.

Wealth Management Intern

DBS Bank

Sep 2018 – May 2019

Singapore, SG

- Performed quantitative operational analyses to identify system inefficiencies and recommend structural improvements.
- Ensured regulatory compliance through detailed validation of Accredited Investor documentation and data controls.

RESEARCH

- Advanced Statistical Modeling for High-Dimensional Personality Data** | *Dr. Fan Zengyan*Oct 2025 – Present
- Engineered preprocessing pipelines to structure high-dimensional psychometric data for multivariate statistical and machine learning analysis.
 - Architected multi-stage frameworks integrating PCA, LASSO regularization, and clustering to identify latent structure and condition-specific interaction effects.
 - Investigated non-linear dimensionality reduction and hybrid modeling strategies to capture geometric and topological structure within feature space.
 - Benchmarked regularized linear models and ensemble tree-based methods (XGBoost), evaluating stability, sparsity, and predictive generalization.

- Multivariate Time Series Forecasting for Equity Price Prediction** | *Dr. Zhou Tianyi*Jan 2025 – Nov 2025
- Developed a multi-horizon return forecasting framework (1-, 5-, 20-day) across NYSE-listed equities.
 - Conducted systematic comparisons of ARIMA, LSTM, and Transformer-based architectures (PatchTST, N-HiTS) under univariate and multivariate formulations.
 - Constructed ensemble-based feature selection pipelines using XGBoost (RFE, permutation importance) to isolate economically meaningful technical signals.
 - Integrated model interpretability techniques to quantify directional sensitivities and multivariate dependencies.
 - Implemented walk-forward backtesting to assess temporal robustness and out-of-sample stability.

- Data-Driven Framework for Enhancing Supply Chain Resilience** | *Dr. Liu Ning*Nov 2024 – Aug 2025
- Designed end-to-end deep learning optimization architectures for inventory control, benchmarking against Predict-Then-Optimize paradigms.
 - Implemented dilated temporal CNN and Multi-Quantile RNN models to capture long-range dependencies and probabilistic demand distributions.
 - Developed stochastic simulation environments to generate synthetic multi-SKU inventory data under variable lead times and demand regimes.
 - Demonstrated cost reductions relative to classical heuristics (Fixed-Order-Quantity, Order-Up-To) through direct cost-function optimization.
 - Documented methodology and quantitative evaluation in a formal technical report for research dissemination.

COMPETENCY

Languages & Scripting: Python, R, SQL, LaTeX
Development & Tools: Git, VS Code, Visual Studio, Jupyter
Spoken Languages: English (Fluent), Mandarin (Native)